Chapter 15

Identical Particles & Electronic Structure

15.1 Introduction

Thus far, we have considered either single-particle systems or systems in which the constituent particles can be distinguished from one another. Nature, however, contains many systems composed of *identical particles*. Every electron has exactly the same mass, charge, and spin as every other electron; the same is true of protons, photons, and many other particles. Unlike classical objects, identical quantum particles possess no hidden serial numbers that allow us to track their individual identities. This fundamental indistinguishability requires us to reconsider how multiparticle states are represented and leads to some of the most important consequences of quantum mechanics, including the distinction between bosons and fermions and the Pauli exclusion principle.

We will apply these concepts to the *electronic structure of solids*. The Hamiltonian determines which single-electron states and energies are available, while Fermi statistics determine how those states may be occupied. We will first consider the free-electron gas, in which conduction electrons move through a uniform background, and then the tight-binding model, in which electrons begin in localized atomic orbitals and acquire an amplitude to move between neighboring atoms. These models illustrate how the behavior of identical fermions, together with the atomic structure of a material, gives rise to Fermi energies, degeneracy pressure, and electronic energy bands.

15.2 The Principle of Indistinguishability

Particles are said to be identical when they belong to the same species and possess no observable intrinsic property by which one can be distinguished from another. It follows that exchanging the labels assigned to identical particles cannot produce an observable change. This is the principle of indistinguishability:

> Dynamical states that differ only by a permutation of identical particles cannot be

distinguished by any observation whatsoever.

This principle does not say that two identical particles must occupy the same state, nor does it say that they cannot be distinguished approximately by their locations. If one electron is localized near the Earth and another near the Moon, it is useful to speak of "the electron near the Earth" and "the electron near the Moon." These descriptions identify spatial states, however, not intrinsic particle identities. If the two electrons exchanged places, with all observable properties exchanged along with them, there would be no experiment that could reveal that an exchange had occurred.

Consider a system of just two identical particles. Any arbitrary vector in the two-particle state space can be expressed as a linear combination of tensor product states as

$$|\Psi\rangle = \sum_{\alpha,\beta} c_{\alpha\beta} \, |\alpha\rangle \, |\beta\rangle$$

where $|\alpha\rangle \, |\beta\rangle = |\alpha\rangle \otimes |\beta\rangle$. In this notation, recall that $|\alpha\rangle$ refers to the first particle and $|\beta\rangle$ refers to the second. A wave function in the two-particle space has the form

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Big(\langle \mathbf{r}_1| \otimes \langle \mathbf{r}_2| \Big) \, |\Psi\rangle = \sum_{\alpha,\beta} c_{\alpha\beta} \, \langle \mathbf{r}_1|\alpha\rangle \, \langle \mathbf{r}_2|\beta\rangle$$

Let P_{12} denote a transposition operator; a transposition is a permutation that interchanges exactly two elements of a set while leaving all other elements fixed. Define P_{12} by the relation

$$P_{12} \, |\alpha\rangle \, |\beta\rangle = |\beta\rangle \, |\alpha\rangle \, .$$

The operator P_{12} is (i) its own inverse, (ii) Hermitian, and (iii) unitary. Its effect on a two-particle wave function is

$$P_{12}\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

As you can see, the transposition operator simply exchanges the labels of the particles. Consider the effect of the exchange operator on the state $|\mathbf{r}_1; +\rangle \, |\mathbf{r}_2; -\rangle$, which represents a spin-up particle at location $\mathbf{r}_1$ and a spin-down particle at location $\mathbf{r}_2$. The exchange operator returns the state $|\mathbf{r}_2; -\rangle \, |\mathbf{r}_1; +\rangle$. In this example, there is still a particle near $\mathbf{r}_1$ that is spin-up and another near $\mathbf{r}_2$ that is spin-down. This example was chosen carefully to make the following point: *if the two particles are truly identical, the labels 1 and 2 have no direct physical meaning.* Interchanging them cannot produce an observable change.

Let A be an operator that represents a dynamical observable and let $|\Psi\rangle$ represent a state of a system of identical particles. The state obtained by interchanging particles i and j is described by the state $P_{ij} \, |\Psi\rangle$, and according to the principle of indistinguishability,

$$\langle \Psi | P_{ij}^{\dagger} \, A \, P_{ij} | \Psi \rangle = \langle \Psi | A | \Psi \rangle \, . \tag{15.1}$$

This must hold for any $|\Psi\rangle$, and since both A and $P_{ij}^\dagger\, A\, P_{ij}$ are Hermitian, it follows that

$$P_{ij}^\dagger A P_{ij} = A, \quad \text{or} \quad [P_{ij}, A] = 0. \tag{15.2}$$

Since A represents an arbitrary observable, we have shown that all physical observables must be permutation-invariant. In a system of two identical particles, P_{12} is the only nontrivial permutation operator, and since $[A, P_{12}] = 0$, the operators A and P_{12} possess a complete set of common eigenvectors. Let $|\Phi\rangle$ denote an eigenvector of the permutation P_{12}, i.e., $P_{12}\,|\Phi\rangle = \lambda\,|\Phi\rangle$, where λ is a real number since P_{12} is Hermitian. Applying P_{12} to the eigenvalue equation, we find

$$P_{12}^2\,|\Phi\rangle = \lambda P_{12}\,|\Phi\rangle = \lambda^2\,|\Phi\rangle \tag{15.3}$$

Since $P_{12}^2 = \mathbb{1}$, it follows that $\lambda = \pm 1$, and we call the eigenstates *symmetric* or *antisymmetric* under the interchange of two identical particles:

$$P_{12}\,|\psi_S\rangle = |\psi_S\rangle\,, \quad \text{and} \quad P_{12}\,|\psi_A\rangle = -\,|\psi_A\rangle \tag{15.4}$$

where $|\psi_S\rangle$ is a symmetric state and $|\psi_A\rangle$ is antisymmetric. A physical observable cannot map a symmetric state into an antisymmetric state or vice versa. Using $[P_{12}, A] = 0$,

$$P_{12}A\,|\psi_S\rangle = AP_{12}\,|\psi_S\rangle = A\,|\psi_S\rangle\,, \quad \text{and} \quad P_{12}A\,|\psi_A\rangle = AP_{12}\,|\psi_A\rangle = -A\,|\psi_A\rangle\,,$$

so $A\,|\psi_S\rangle$ remains symmetric and $A\,|\psi_A\rangle$ remains antisymmetric.

The Hamiltonian is a physical observable, so it must also be invariant under permutations. Consequently, exchange symmetry is conserved by the time evolution. A state that is initially symmetric remains symmetric, and a state that is initially antisymmetric remains antisymmetric. A typical two-particle Hamiltonian has the form

$$H = \frac{\mathbf{p}_1^2}{2m} + \frac{\mathbf{p}_2^2}{2m} + V_0(\mathbf{r}_1) + V_0(\mathbf{r}_2) + V(\mathbf{r}_1, \mathbf{r}_2)$$

Under the exchange $\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$, $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$, H remains unchanged provided $V(\mathbf{r}_1, \mathbf{r}_2) = V(\mathbf{r}_2, \mathbf{r}_1)$, which is automatically satisfied by any central potential $V(\mathbf{r}_1, \mathbf{r}_2) = V\left(|\mathbf{r}_2 - \mathbf{r}_1|\right)$.

15.3 The Symmetrization Postulate

The principle of indistinguishability requires observable physics to be invariant under permutations, but it does not, by itself, require every allowed state vector to be symmetric or antisymmetric. The stronger restriction on the allowed states is supplied by the symmetrization postulate:

> The states of a system of identical particles are either completely symmetric or completely antisymmetric under the interchange of any pair of particles.

In nature, the exchange symmetry of a system of identical particles is determined by the spin of the particles. Identical particles with integer spin occupy symmetric states and are called **bosons**, whereas identical particles with half-integer spin occupy antisymmetric states and are called **fermions**. This relationship is known as the **spin-statistics connection**. In relativistic quantum field theory, it follows from the spin-statistics theorem; in nonrelativistic quantum mechanics, it must be introduced as an empirical fact.

Example 15.1: Two Identical Particles in a Box

The symmetrization postulate applies to the complete state, including both spatial and spin degrees of freedom. If the state factorizes as

$$|\Psi\rangle = |\psi\rangle_{\text{spatial}} \, |\chi\rangle_{\text{spin}} \, ,$$

then the exchange symmetries of the two factors must combine to give the symmetry of the complete state. In this example, suppose the spin state is symmetric, i.e.,

$$P_{12} \, |\chi\rangle_{\text{spin}} = + \, |\chi\rangle_{\text{spin}} \, .$$

The spatial state must then be symmetric for bosons and antisymmetric for fermions:

$$P_{12} \, |\psi\rangle_{\text{spatial}} = \begin{cases} + \, |\psi\rangle_{\text{spatial}} \, , & \text{bosons,} \\ - \, |\psi\rangle_{\text{spatial}} \, , & \text{fermions.} \end{cases}$$

We will suppress the unchanged spin factor and work only with the spatial wave function.

Example: Suppose we have two noninteracting particles, both of mass m, in an infinite square well. The one-particle states are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad E_n = n^2 K,$$

where $K = \pi^2\hbar^2/2ma^2$. If the particles are distinguishable, the composite wave function is

$$\psi_{n_1,n_2}(x_1, x_2) = \psi_{n_1}(x_1)\psi_{n_2}(x_2), \quad E_{n_1,n_2} = (n_1^2 + n_2^2)K.$$

For example, the ground state is

$$\psi_{11} = \frac{2}{a} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right), \quad E_{11} = 2K,$$

and the first excited state is two-fold degenerate:

$$\psi_{12} = \frac{2}{a}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right), \quad E_{12} = 5K,$$

$$\psi_{21} = \frac{2}{a}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right), \quad E_{21} = 5K,$$

If, however, the two particles are identical bosons, the ground state is unchanged, but the first excited state is nondegenerate:

$$\psi_S = \frac{\sqrt{2}}{a}\left[\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) + \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right], \quad E_S = 5K.$$

If the particles are identical fermions, there is no state with energy $2K$; the ground state is

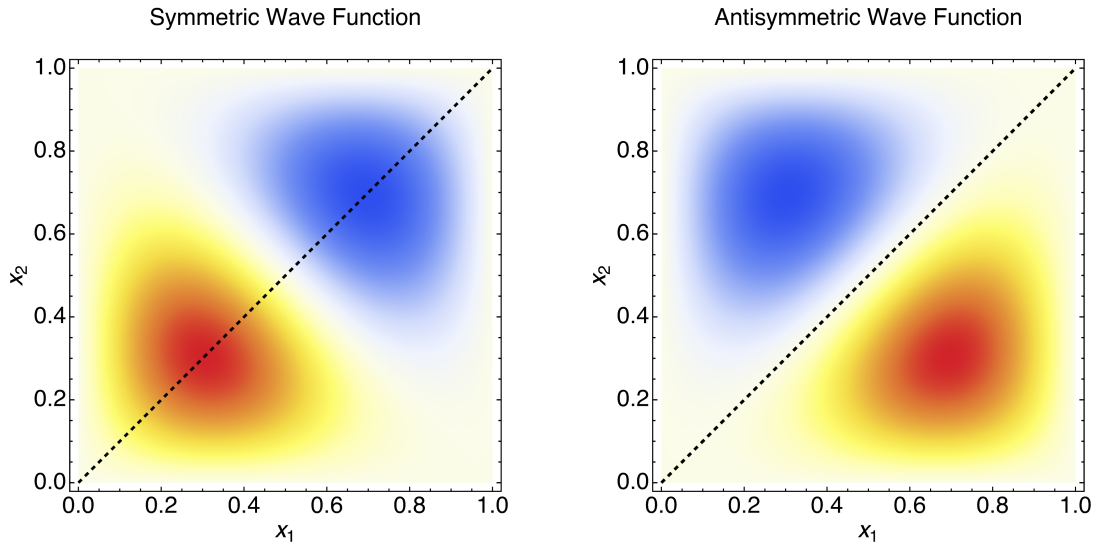
$$\psi_A = \frac{\sqrt{2}}{a}\left[\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right], \quad E_A = 5K.$$

There is no state with energy $2K$ because such a state would require both particles to occupy the one-particle ground state, $n_1 = n_2 = 1$. Antisymmetrizing the product state gives

$$\psi_{11}^{(A)}(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\psi_1(x_1)\psi_1(x_2) - \psi_1(x_2)\psi_1(x_1)\right] = 0$$

This is an example of the **Pauli exclusion principle**: two identical fermions cannot occupy the same complete one-particle state. In the present example, their spin state is symmetric, so placing both particles in the same spatial state would make the complete state vanish. If the two fermions instead had an antisymmetric spin state, they could occupy the same spatial state because the complete state would still be antisymmetric.

The figure below shows the symmetric and antisymmetric wave functions plotted against x_1 and x_2. The exchange $x_1 \leftrightarrow x_2$ corresponds to reflection across the dashed line $x_1 = x_2$. The symmetric wave function is unchanged by this reflection; points mirrored across the diagonal have the same value and therefore the same color. The antisymmetric wave function changes sign; regions of positive amplitude on one side correspond to regions of negative amplitude on the other. Consequently, the antisymmetric wave function vanishes along $x_1 = x_2$, producing the nodal line visible in the plot. Thus, for this state, the two fermions have zero probability of being found at the same position.



Mean-square separation

We might try to characterize the locations of the particles by calculating $\langle x_1 \rangle$ and $\langle x_2 \rangle$. For both the symmetric and antisymmetric states, we find $\langle x_1 \rangle = \langle x_2 \rangle = a/2$. Thus, if we temporarily retain the artificial particle labels, the average position associated with each label is the center of the well. Strictly speaking, however, x_1 and x_2 are not individually physical observables for a system of identical particles. Under exchange,

$$P_{12}^\dagger x_1 P_{12} = x_2, \qquad P_{12}^\dagger x_2 P_{12} = x_1,$$

so neither operator commutes with P_{12}. The sum of the positions is permutation-invariant:

$$P_{12}^\dagger (x_1 + x_2) P_{12} = x_1 + x_2.$$

It is therefore a physical observable, and its expectation value is $\langle x_1 + x_2 \rangle = a$. Equivalently, the center-of-mass coordinate $X = (x_1 + x_2)/2$ is physical and satisfies $\langle X \rangle = a/2$. The signed difference $x_1 - x_2$ is not permutation-invariant:

$$P_{12}^\dagger (x_1 - x_2) P_{12} = -(x_1 - x_2).$$

Its sign depends on which identical particle was arbitrarily assigned the label 1. Nevertheless, its expectation value vanishes in both states:

$$\langle x_1 - x_2 \rangle_S = \langle x_1 - x_2 \rangle_A = 0.$$

This simply reflects the absence of any physical distinction between the two particle labels.

The square of the separation is permutation-invariant:

$$P_{12}^{\dagger}(x_1 - x_2)^2 P_{12} = (x_1 - x_2)^2.$$

It is therefore a physical observable. Using

$$(x_1 - x_2)^2 = x_1^2 + x_2^2 - 2x_1x_2,$$

we find

$$\langle x_1^2 + x_2^2 \rangle = a^2 \left(\frac{2}{3} - \frac{5}{8\pi^2} \right)$$

for both exchange symmetries. The correlation term, however, depends on the relative sign in the wave function:

$$\langle x_1 x_2 \rangle_{S/A} = \frac{a^2}{4} \pm \frac{256a^2}{81\pi^4},$$

where the upper sign applies to the symmetric state and the lower sign to the antisymmetric state. Consequently,

$$\big\langle (x_1 - x_2)^2 \big\rangle_S = a^2 \left(\frac{1}{6} - \frac{5}{8\pi^2} - \frac{512}{81\pi^4} \right)$$

and

$$\big\langle (x_1 - x_2)^2 \big\rangle_A = a^2 \left(\frac{1}{6} - \frac{5}{8\pi^2} + \frac{512}{81\pi^4} \right).$$

Numerically,

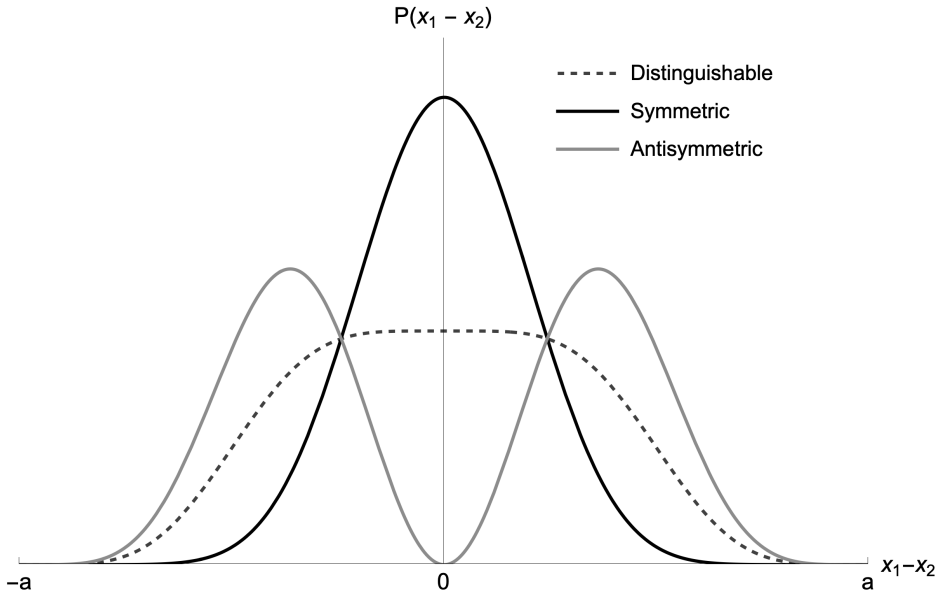
$$\big\langle (x_1 - x_2)^2 \big\rangle_S \approx 0.0384a^2,$$

whereas

$$\big\langle (x_1 - x_2)^2 \big\rangle_A \approx 0.1682a^2.$$

Although the particles do not interact, the antisymmetric state has a much larger mean-square separation than the symmetric state. The two states have the same average center-of-mass position and the same individual one-particle position distribution, but they possess different spatial correlations. Symmetric exchange favors configurations in which the particles are closer together, while antisymmetric exchange suppresses such configurations and forces the wave function to vanish when $x_1 = x_2$.

The system behaves as if there were an attractive force between identical bosons pulling them closer together and a repulsive force between identical fermions pushing them apart. We call it an **exchange force**, but it's not really a force at all. Rather, it is a consequence of the symmetrization requirement, which is purely quantum-mechanical with no classical analog. The exchange force is best understood as a statistical correlation, not a dynamical interaction.



Probability density for the signed relative coordinate $x_1 - x_2$, for two particles in the first excited configuration of an infinite square well of width a, with one particle in each of the $n = 1$ and $n = 2$ states. The dashed curve is for distinguishable particles; the black and gray curves show, respectively, symmetric and antisymmetric spatial states.

Including spin states

Thus far, we assumed the two particles had a symmetric spin state. Let's remove that restriction and determine all allowed states for two identical spin-1/2 particles. The particles are fermions, so the complete state must be antisymmetric. Because the Hamiltonian in this example is spin-independent, we can choose a basis of energy eigenstates that factor into spatial and spin parts.

The four spin states of two spin-1/2 particles consist of one antisymmetric singlet and three symmetric triplet states. The antisymmetric singlet is

$$|0,0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle - |-+\rangle \Big),$$

and the three symmetric states comprising the triplet are

$$|1,1\rangle = |++\rangle\,,$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle + |-+\rangle \Big),$$

$$|1,-1\rangle = |--\rangle\,.$$

It is interesting that the coupled spin states emerge naturally from the symmetrization requirement. Using ket notation for the spin state, the nondegenerate ground state is

$$|\Psi_0(x_1, x_2)\rangle = \psi_1(x_1)\psi_1(x_2)\,|0,0\rangle\,, \quad E_{11} = 2K$$

The antisymmetric spin singlet allows the two fermions to share the same lowest-energy spatial orbital without violating the Pauli exclusion principle.

At the first-excited energy level, $n_1 = 1$ and $n_2 = 2$, so $E_{12} = 5K$. The symmetric spatial state must be paired with the antisymmetric singlet state:

$$\frac{1}{\sqrt{2}}\Big[\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)\Big]\;|0,0\rangle\,.$$

The antisymmetric spatial state must be paired with one of the three symmetric triplet states:

$$\frac{1}{\sqrt{2}}\Big[\psi_1(x_1)\psi_2(x_2) - \psi_2(x_1)\psi_1(x_2)\Big]\;|1,m\rangle\,, \quad m = -1, 0, 1$$

15.4 The Free Electron Gas

Consider N noninteracting electrons confined to a cubic box of side length L and volume $V = L^3$. The walls are impenetrable, so the single-electron spatial wave functions vanish at the boundaries. The normalized single-particle wave functions are

$$\psi_{n_x,n_y,n_z} = \sqrt{\frac{8}{V}}\,\sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right) \tag{15.5}$$

where $n_x, n_y, n_z = 1, 2, 3, \ldots$ The corresponding energies are

$$E_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2mL^2}\left(n_x^2 + n_y^2 + n_z^2\right) = \frac{\hbar^2 k^2}{2m} \tag{15.6}$$

where k is the magnitude of the wave vector

$$\mathbf{k} = \frac{n_x\pi}{L}\hat{\mathbf{x}} + \frac{n_y\pi}{L}\hat{\mathbf{y}} + \frac{n_z\pi}{L}\hat{\mathbf{z}} \tag{15.7}$$

The spacing between adjacent values of k_i is $\Delta k_i = \pi/L$, so each spatial state occupies the k-space volume $\Delta^3 k = \pi^3/V$. Because n_i are positive, the allowed points lie in the positive octant of k-space.

At zero temperature, the electrons fill the lowest energy one-particle states. Since electrons are spin-1/2, they cannot all occupy the same lowest energy state; each spatial state can accommodate two electrons, one spin-up and the other spin-down. The occupied states fill the positive-octant portion of a sphere in k-space of radius k_F, called the Fermi wave vector. The total number of

electrons that occupy states inside this region is approximately

$$N = 2 \times \frac{\frac{1}{8}\left(\frac{4}{3}\pi k_F^3\right)}{\pi^3/V} = \frac{V k_F^3}{3\pi^2} \tag{15.8}$$

The occupied states do not form an exact spherical octant because the allowed points are discrete. The sphere-counting argument becomes accurate in the thermodynamic limit, where boundary corrections are negligible. Therefore,

$$\boxed{k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} = \left(3\pi^2 n\right)^{1/3}} \tag{15.9}$$

where $n = N/V$ is the free electron number density (the number of free electrons per unit volume). The Fermi energy is the energy of the highest occupied one-electron state at zero temperature:

$$\boxed{E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}} \tag{15.10}$$

Both k_F and E_F are determined by the electron number density $n = N/V$. For a particular metal under ordinary conditions, the conduction-electron density is approximately fixed by its atomic density and the number of conduction electrons contributed by each atom. For example, the density of copper is 8.96 g/cm^3, and its atomic weight is 63.5 g/mol. If we assume that each atom contributes one conduction electron, then copper has an electron number density of 8.49×10^{28} m^{-3} and a corresponding Fermi energy of about 7.0 eV.

The total ground-state energy of the electron gas may be calculated using the one-electron density of states $g(E)$, defined so that

$$g(E)\, dE$$

is the number of available one-electron states with energies between E and $E + dE$. At zero temperature, every one-electron state below the Fermi energy is occupied, so

$$N = \int_0^{E_F} g(E) dE \tag{15.11}$$

and the total ground-state energy is

$$U = \int_0^{E_F} E\, g(E) dE \tag{15.12}$$

The number of electrons occupying single-particle states within a shell of thickness dk is

$$dN = \frac{2 \times \frac{1}{8}\left(4\pi k^2 dk\right)}{\pi^3/V} = \frac{V}{\pi^2} k^2 dk$$

We can find the density of states $g(E) = dN/dE$ for the hard-wall electron gas using the chain rule:

$$dN = g(E)dE = g(E)\frac{dE}{dk}dk,$$

where

$$E = \frac{\hbar^2 k^2}{2m} \Rightarrow \frac{dE}{dk} = \frac{\hbar^2 k}{m}.$$

Comparing our two expressions for dN, we find

$$g(E)\frac{\hbar^2 k}{m} = \frac{V}{\pi^2}k^2$$

and using $k = \sqrt{2mE}/\hbar$, we obtain

$$\boxed{g(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}} \tag{15.13}$$

The total ground-state energy is therefore

$$U = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^{E_F} E^{3/2}\,dE$$

$$= \frac{V}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{5/2}$$

Using the expression for N in terms of the Fermi energy,

$$\boxed{U = \frac{3}{5}NE_F,} \tag{15.14}$$

which means that the average energy per electron is $\frac{3}{5}E_F$. It makes sense that the average energy would be less than the Fermi energy since it represents the highest available single-electron energy.

If we hold the number of electrons fixed, the ground-state energy of the gas decreases as the box expands. The electron gas exerts an outward pressure on the walls of the container, much like an ordinary gas. In this case, however, the temperature is zero, so the pressure is not caused by thermal agitation. The Pauli exclusion principle forces electrons to occupy high-energy states, producing a nonzero kinetic energy and pressure even at zero temperature. This purely quantum-mechanical pressure is called **degeneracy pressure**. To calculate it, recall that when a gas expands quasistatically by an amount dV, the work it performs is

$$dW = P\,dV.$$

This work comes at the expense of the gas's internal energy, so

$$dU = -P\,dV.$$

Therefore, at fixed particle number,

$$P = -\left(\frac{\partial U}{\partial V}\right)_N,$$

and the degeneracy pressure of the free-electron gas can be written as

$$\boxed{P = \frac{2}{3}\frac{U}{V} = \frac{2}{5}nE_F} \tag{15.15}$$

Interestingly, the expression $P = 2U/(3V)$ is the same as that for an ideal gas, where U is the internal energy of the gas at temperature T. This relation holds for both a classical ideal gas and a noninteracting quantum gas because the one-particle energy has the quadratic form $E \propto p^2$. At $T = 0$, a classical ideal gas would have $U = P = 0$, whereas the "Fermi gas" retains kinetic energy as a result of the Pauli exclusion principle. Using copper as an example, the degeneracy pressure comes out to 3.8×10^{10} Pa or 38 GPa. Degeneracy pressure makes a large contribution to copper's resistance to compression, but it's not the whole explanation. Core-electron Pauli repulsion, electrostatic interactions, and the band structure supply most of the remaining stiffness.

15.5 The Tight-Binding Model

The free-electron gas provides a useful first approximation to the electrons in a metal. In that model, the positively charged ions are replaced by a uniform background, and the conduction electrons move almost freely throughout the solid. The Pauli exclusion principle determines how the available one-electron states are filled, leading to the Fermi energy and degeneracy pressure. What the model leaves out, however, is the discrete atomic structure of the solid. A real crystal consists of atoms arranged in a *periodic lattice*, and the electrons experience the potential of each atomic nucleus together with the fields produced by the other electrons.

We will now approach the problem of electronic structure from a complementary limit using the *tight-binding model*. Imagine beginning with a collection of widely separated atoms. Each atom possesses localized electronic states with discrete energies. If the atoms are brought together to form a crystal, neighboring atomic wave functions begin to overlap. An electron initially associated with one atom then acquires some amplitude to move, or "hop," to a neighboring atom. The resulting energy eigenstates are no longer localized on individual atoms but extend throughout the lattice.

For simplicity, consider a one-dimensional lattice of N identical atoms separated by a distance a, with the nth atom located at $x_n = na$. When the electron sits on the nth atom in isolation, it occupies the state $|n\rangle$ with energy E_0 where $n = 0, 1, \ldots, N-1$. We take these states to be orthonormal; more refined treatments include orbital overlap. In addition to the on-site energy, we

also include nearest-neighbor hopping terms that describe an electron hopping from one atom to neighboring atoms. The tight-binding Hamiltonian is

$$H = E_0 \sum_n |n\rangle \langle n| - t \sum_n \Big(|n\rangle \langle n+1| + |n+1\rangle \langle n| \Big) \tag{15.16}$$

where $t > 0$ is the hopping parameter. We will also impose periodic boundary conditions such that $|N\rangle = |0\rangle$. This might seem arbitrary, but it's merely a mathematical convenience: it eliminates special boundary sites (edge effects) and makes every lattice site equivalent. Different boundary conditions lead to slightly different states for finite N, but their predictions for bulk properties approach the same result as N becomes large. Periodic boundary conditions should not be confused with the physical periodicity of the lattice.

15.5.1 Discrete Translation Operator

Our lattice has discrete translational symmetry; it's invariant under spatial translation in either direction by the lattice spacing a. Define the discrete translation operator by its action on the localized basis states:

$$T_a |n\rangle = |n+1\rangle \tag{15.17}$$

From Eq. (15.17), it follows that T_a is unitary. Furthermore, the tight-binding Hamiltonian is invariant under discrete translation:

$$T_a H T_a^\dagger = E_0 \sum_n |n+1\rangle \langle n+1| - t \sum_n \Big(|n+1\rangle \langle n+2| + |n+2\rangle \langle n+1| \Big).$$

Relabeling the summation index, we find $T_a H T_a^\dagger = H$ or $[H, T_a] = 0$. Thus, the energy eigenstates can be chosen to be eigenstates of discrete translation. Because T_a is unitary, its eigenvalues must satisfy $|\lambda_k| = 1$. Thus, $\lambda_k = e^{-i\theta_k}$. Furthermore, for an eigenstate $|k\rangle$ of the translation operator,

$$T_a^n |k\rangle = e^{-in\theta_k} |k\rangle$$

and thus $|k\rangle$ is an eigenstate with eigenvalue $e^{-in\theta_k}$. Since the operator T_a^n corresponds to translation by na, we define the "wave number" $k = \theta_k/a$ such that

$$T_a^n |k\rangle = e^{-ikna} |k\rangle \tag{15.18}$$

We'll call $|k\rangle$ a momentum eigenstate; more precisely, k is called the crystal wave number or Bloch wave number, while $\hbar k$ is the crystal momentum. The momentum eigenstate can be expanded in terms of localized states as

$$|k\rangle = \sum_m c_m |m\rangle \tag{15.19}$$

Substituting Eq. (15.19) into Eq. (15.18) with $n = 1$ gives

$$\sum_m c_m \left|m+1\right\rangle = \sum_m c_m e^{-ika} \left|m\right\rangle$$

Relabel the summation index on the left-hand side $m + 1 \to m$ such that

$$\sum_m c_{m-1} \left|m\right\rangle = \sum_m c_m e^{-ika} \left|m\right\rangle$$

Comparing coefficients, $c_{m-1} = c_m e^{-ika}$ or $c_m = e^{ika} c_{m-1}$, which is a recursion relation. Repeated application of the recursion gives

$$c_m = c_0 e^{ikma}$$

The momentum eigenstate is

$$\left|k\right\rangle = c_0 \sum_n e^{ikna} \left|n\right\rangle$$

and c_0 can be fixed by normalization:

$$\langle k|k\rangle = |c_0|^2 \sum_{n,m} e^{ik(n-m)a} \delta_{nm} = |c_0|^2 \sum_{n=0}^{N-1} 1 = |c_0|^2 N = 1$$

Hence, the momentum eigenstates are

$$\left|k\right\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} \left|n\right\rangle \tag{15.20}$$

Each state $|n\rangle$ is localized near the nth lattice site. By contrast, each momentum eigenstate $|k\rangle$ is a coherent superposition extending over the entire lattice, in which the amplitude on each site differs from that on the preceding site by the same phase factor e^{ika}. The state is therefore delocalized but possesses a definite spatial phase pattern across the lattice.

15.5.2 Allowed Wave Numbers & Energies

The periodic boundary conditions restrict the possible values of k. Translating through all N sites returns the state to its original site, so it must be the case that

$$T_a^N = \mathbb{1} \qquad \text{(periodic boundary conditions)}$$

Thus,

$$T_a^N \left|k\right\rangle = e^{-ikNa} \left|k\right\rangle = \left|k\right\rangle$$

which requires

$$e^{-ikNa} = 1.$$

This is a transcendental equation with the solution

$$kNa = 2\pi j, \quad j = 0, 1, 2, \ldots, N-1.$$

The allowed wave numbers are

$$k_j = \frac{2\pi j}{Na}, \quad j = 0, 1, 2, \ldots, N-1. \tag{15.21}$$

and the momentum eigenstates are

$$\boxed{|k_j\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{i2\pi jn/N} |n\rangle} \tag{15.22}$$

They form the discrete Fourier transform of the localized basis. Their orthogonality follows from

$$\langle k_{j'} | k_j \rangle = \frac{1}{N} \sum_{n=0}^{N-1} e^{i(k_j - k_{j'})na} = \delta_{jj'}.$$

In this model, each allowed translation eigenvalue corresponds to a one-dimensional eigenspace. Since $[H, T_a] = 0$,

$$T_a H |k\rangle = H T_a |k\rangle = e^{-ika} H |k\rangle .$$

Thus, $H|k\rangle$ belongs to the same one-dimensional translation eigenspace as $|k\rangle$, so it must be proportional to $|k\rangle$. Therefore, $|k\rangle$ is also an energy eigenstate. We can use the energy eigenvalue equation to determine the allowed energies of the system:

$$H |k\rangle = c_0 \sum_n e^{ikna} \Big(E_0 |n\rangle - t |n-1\rangle - t |n+1\rangle \Big)$$

Relabeling the sums in the hopping terms gives

$$\begin{aligned} H |k\rangle &= c_0 \sum_n e^{ikna} \Big(E_0 |n\rangle - te^{ika} |n\rangle - te^{-ika} |n\rangle \Big) \\ &= \Big[E_0 - t(e^{ika} + e^{-ika}) \Big] |k\rangle \\ &= [E_0 - 2t\cos(ka)] |k\rangle \end{aligned}$$

Hence, the allowed energies are

$$\boxed{E_j = E_0 - 2t\cos\left(\frac{2\pi j}{N}\right), \quad j = 0, 1, 2, \ldots, N-1.} \tag{15.23}$$

We may refer to the expression $E(k) = E_0 - 2t\cos(ka)$ as well.

15.5.3 The Brillouin Zone

The wave number k is not unique; any k that differs by an integer multiple of $2\pi/a$ is associated with the same momentum eigenstate. Indeed, for any integer ℓ,

$$|k + 2\pi\ell/a\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} e^{i2\pi\ell n} |n\rangle = |k\rangle$$

Thus, the wave numbers k and $k + 2\pi\ell/a$ belong to the same equivalence class. We may therefore choose one interval of width $2\pi/a$ containing one representative of each equivalence class. The conventional choice is the **first Brillouin zone**:

$$\boxed{-\frac{\pi}{a} \leq k < \frac{\pi}{a}}$$

The Brillouin zone does not impose an additional physical restriction on the electron. It simply avoids labeling the same crystal state more than once. For a finite chain, the first Brillouin zone contains N allowed values of k, separated by

$$\Delta k = \frac{2\pi}{Na}.$$

As N becomes macroscopically large, these values become extremely closely spaced and k may be treated as continuous.

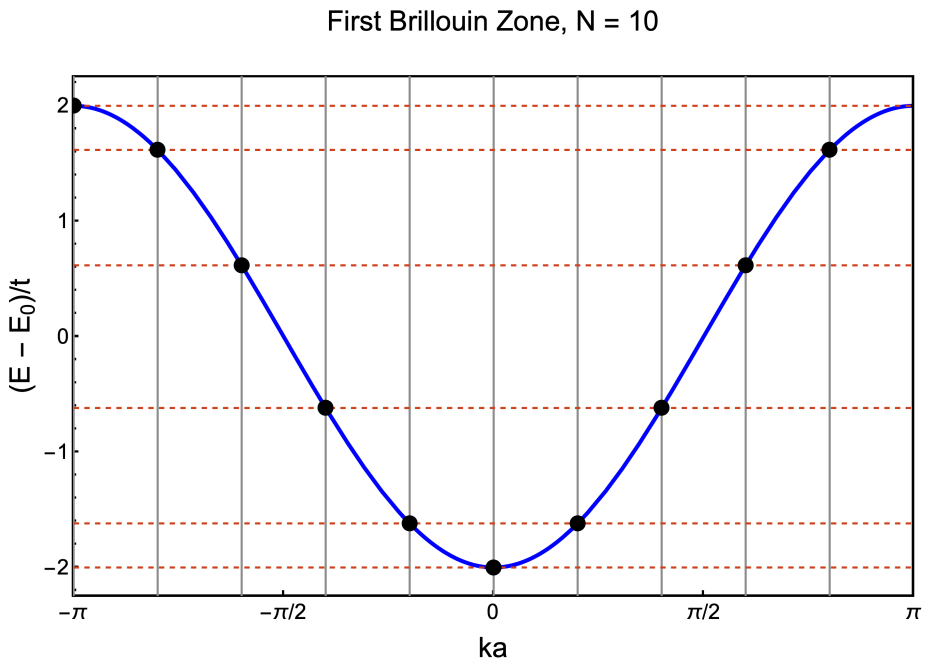
When the hopping is neglected, every localized state $|n\rangle$ has the same energy E_0. The energy E_0 is therefore N-fold degenerate. Turning on the hopping term couples the localized states and distributes the N crystal states over the energies

$$E(k) = E_0 - 2t\cos(ka)$$

Since $-1 \leq \cos(ka) \leq 1$, the allowed energies satisfy

$$E_0 - 2t \leq E(k) \leq E_0 + 2t.$$

In other words, the original atomic energy level has broadened into an *energy band* of width $4t$. Strictly speaking, the finite crystal still has N discrete spatial states. For macroscopic N, however, the energy levels are so closely spaced that they appear to form a continuous band. The spectrum is shown in the figure below for $N = 10$ and $t > 0$.



In summary, when $t = 0$, the localized states $|n\rangle$ are degenerate energy eigenstates with energy E_0. Adding any nonzero hopping term, no matter how small, couples neighboring sites, and the energy eigenstates become delocalized Bloch states $|k\rangle$. In each Bloch state, the electron has the same probability of being found at any atom and the probability amplitude acquires a consistent phase difference ka from one site to the next. At the same time, hopping lifts the original N-fold degeneracy and spreads the single atomic energy E_0 into a band of width $4t$. As $t \to 0$, the band becomes arbitrarily narrow, but the Bloch states remain delocalized for every $t \neq 0$; only at exactly $t = 0$ can the localized states themselves be chosen as energy eigenstates.

15.5.4 Effective Mass

For small crystal momentum, $ka \ll 1$ and $\cos(ka) \approx 1 - (ka)^2/2$. The energy is therefore

$$E(k) \approx E_0 - 2t + ta^2k^2 = E_{\min} + \frac{\hbar^2 k^2}{2m^*}$$

where

$$m^* = \frac{\hbar^2}{2ta^2}$$

is the *effective mass* near the bottom of the band. In terms of the crystal momentum $p_{\text{cr}} = \hbar k$,

$$E(k) \approx E_{\min} + \frac{p_{\text{cr}}^2}{2m^*}$$

Thus, an electron near the bottom of the energy band responds approximately like a free particle with energy $p^2/2m$, but with the bare electron mass replaced by an effective mass determined by

the curvature of the band. More generally,

$$\frac{1}{m^*(k)} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{2ta^2}{\hbar^2}\cos(ka).$$

15.5.5 Filling the Band with Electrons

We now place N_e noninteracting electrons in the lattice at zero temperature. Since the Hamiltonian is spin independent, each spatial state $|k\rangle$ has two associated spin states; one spin up and the other spin down. The band contains N spatial one-electron states and can therefore accommodate $2N$ electrons. The many-electron ground state is the antisymmetrized state obtained by filling the lowest-energy spin orbitals, with no more than one electron in each spin orbital.

At zero temperature, the electrons occupy the lowest available energies up to the Fermi energy. If a band is partially filled, the Fermi energy lies within it, so occupied states are separated from unoccupied states by arbitrarily small energies. An applied electric field can then shift the distribution of occupied crystal momenta, producing an imbalance between electrons moving in opposite directions and hence an electrical current. A partially filled band therefore describes a conductor in the noninteracting band picture.

A completely filled band behaves differently. Every state in the band is occupied, so the Pauli exclusion principle prevents electrons from rearranging themselves within the band in response to a weak electric field. Furthermore, the currents associated with states of opposite crystal momentum cancel. If the filled band is separated from the next unoccupied band by an energy gap, a weak field cannot promote electrons into that band either. The material is then a **band insulator**. In our tight-binding model, there is no energy gap because there's only one energy band.

A real crystal generally possesses many energy bands. The highest completely filled band is called the **valence band**, while the next higher band is called the **conduction band**. In an insulator or semiconductor, these bands are separated by an energy gap. A semiconductor has a relatively small gap, so thermal energy or absorbed light can excite electrons from the valence band into the conduction band. The excited electrons can then move through the conduction band and carry current, while the vacancies they leave behind, called **holes**, can also move through the valence band and contribute to the current. An ideal intrinsic semiconductor is therefore insulating at zero temperature but becomes conducting as electrons are excited across the gap.

15.5.6 Bonding & Antibonding

The tight-binding energy is minimized when $\cos(ka) = 1$, which occurs at $k = 0$. The corresponding state is

$$|k_{\min}\rangle = \frac{1}{\sqrt{N}}\sum_n |n\rangle$$

and its energy is $E_{\min} = E_0 - 2t$. All amplitudes have the same sign. In other words, the amplitudes from the two neighboring sites reinforce the amplitude at site n, and the minus sign in the hopping Hamiltonian lowers the energy. This uniform phase pattern is called the bonding state.

The energy is maximized when $\cos(ka) = -1$, which occurs at the edge of the Brillouin zone where $k = \pm\pi/a$. The maximum-energy state is

$$|k_{\max}\rangle = \frac{1}{\sqrt{N}} \sum_n e^{i\pi n} |n\rangle = \frac{1}{\sqrt{N}} \sum_n (-1)^n |n\rangle$$

and its energy is $E_{\max} = E_0 + 2t$. The amplitudes from the neighboring sites are opposite in phase to the amplitude at site n, so the hopping term raises the energy. This alternating phase pattern is called the antibonding state.

The terms "bonding" and "antibonding" apply most naturally to a real-space interpretation if the states $|n\rangle$ represent localized atomic orbitals $\phi_n(x)$. The corresponding wave function is

$$\psi_k(x) = \frac{1}{\sqrt{N}} \sum_n e^{ikna} \phi_n(x).$$

For $k = 0$, neighboring orbitals enter with the same sign and reinforce one another where they overlap. The resulting bonding wave function varies relatively smoothly through the lattice and has the lowest energy. For $k = \pm\pi/a$, neighboring orbitals enter with opposite signs and tend to cancel in their overlap regions, typically producing additional nodes. The resulting antibonding wave function varies more rapidly in space and has the highest energy.